Package Assignment and Routing System (PARS)

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Introduction

- Suppose you are a warehouse operator serving a city.
- Your job is to distribute goods from the warehouse using delivery trucks to transport them to their final destination in an efficient manner.
- However, a single truck can carry far more than just one delivery at a time.
- As such, we must design efficient routes for these trucks.

Motivation

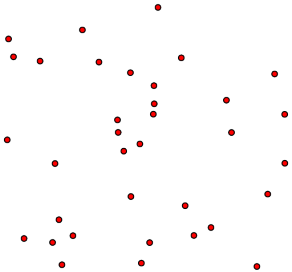
For our project, we'll be using the analogy of delivering packages to homes.

- However, also consider school bus routing, human and robotic warehouse pickers/stockers, waste collection, snow plowing, coordination of emergency services, etc.
- Finding an optimal solution is computationally expensive. As such, we must resort to heuristics to find a reasonable solution in a reasonable amount of time.
- AI revolves around this idea of making good enough decisions in a reasonable amount of time.

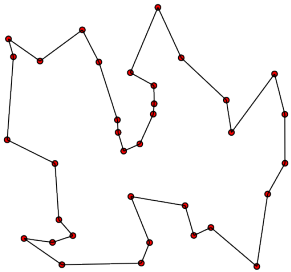
Foundational problem

Traveling salesman problem (TSP) — "Given some origin city, what's the shortest possible route that visits each city exactly once and returns to the origin city?"

- Modeling this as a graph theory problem, we can assume a complete graph K_n where nodes are cities and edge weights are pre-calculated distances between those cities.



Unsolved



Solved

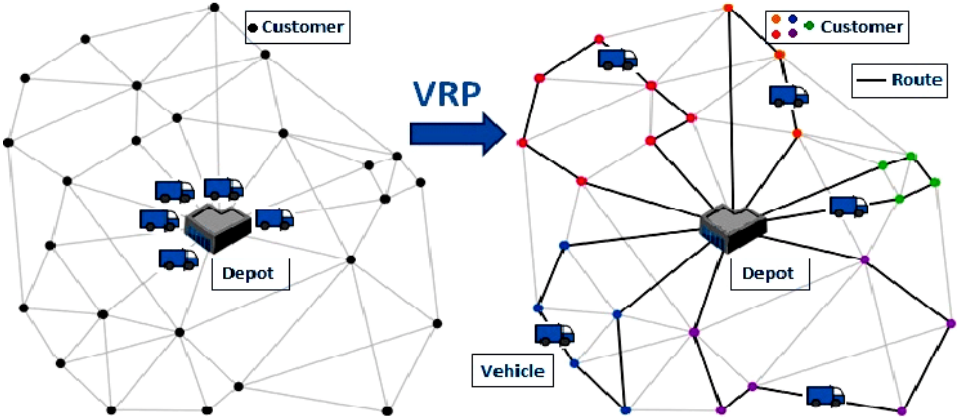
Vehicle routing problem

Vehicle routing problem (VRP) — Generalization of the traveling salesman problem which asks "what is the optimal set of routes for a fleet of vehicles to traverse in order to deliver to a given set of customers?"

- Typically, the goal is to minimize the total distance traveled by all vehicles, however, many more metrics might be considered.
- Can include any number of constraints that must be satisfied—a vehicle can only carry so much weight, a driver can only work for so long.
- Can be thought of as attempting to solve multiple traveling salesman problems or a multi-agent search problem.

VRPs also assume a complete graph where actual road routing is abstracted away. However, we will model our problem to utilize real-world road network information.

VRP overview



Gupta, Ashima & Saini, Sanjay. (2017). An Enhanced Ant Colony Optimization Algorithm for Vehicle Routing Problem with Time Windows. 267-274. 10.1109/ICoAC.2017.8441175.

Modern solutions

Genetic Algorithms (GA) — Technique inspired by natural selection. Here the best routes are selected and more optimal routes are crossed with each other. Mutations are introduced to avoid getting stuck in local optimas.

Ant colony optimization (ACO) — Technique which emulates pheromones used by ants to find paths. Virtual ants leave a chemical trail. When an ant completes an efficient route, it lays down a strong pheromone trail. The next ant is more likely to choose a path with high pheromone levels. Over time, trails on inefficient routes evaporate.

Simulated annealing (SA) — Emulates the cooling of hot metal. Early on, the algorithm accepts longer routes to "see where they might go". Though, as we cool, we become more conservative. Good for finding global optimums without getting stuck in a local optima.

Formal definition

Let $G = (V, E)$ be a connected, weighted graph representing a road network where

- vertices represent intersecting roads and/or customers,
- vertices also possess latitude/longitude information (x, y),
- all trucks start and end at a central warehouse described by a node $w \in V$,
- $C \subseteq V \setminus \{w\}$ is the set of destinations (customers) the trucks must visit,
- edges represent the existence of a road connecting two vertices where edge weights describe the distance between the two nodes.

Let T be a finite set of trucks each with fixed capacity k where the total customers a single truck visits cannot exceed k.

We define function $\text{PARS}(G, w, C, T, k)$ which outputs a mapping $T \to \{(w, \ldots, w) \subseteq V, \ldots\}$ which describes the route of each truck starting at w and ending at w. The goal of this function is to minimize the total distance traveled, the sum of edge weights, across all trucks, while also maximizing customers visited.

Our proposal

1. Use k-means clustering to first partition the set of customers by proximity in order to not exceed a single truck's capacity.
2. Now solve a traveling salesman problem for each cluster using nearest-neighbor heuristic for choosing the next stop in the route and A* search for routing from one stop to the next.
3. Use reinforcement learning to attempt to find a more optimal solution over our classical approach.
4. We define our reward function to reward visiting as many customers as possible while minimizing overall distance traveled.

Our primary goal is to reduce the distance traveled by all trucks in total. We will record this metric over many simulations, then compare and contrast our classical and modern solutions.

Main reference

I. Bello, H. Pham, Q. Le, M. Norouzi, S. Bengio, "Neural Combinatorial Optimization with Reinforcement Learning," 2017. [Online] Available: `https://arxiv.org/abs/1611.09940`

- This paper introduces a framework for solving combinatorial optimization problems using neural networks and reinforcement learning.
- Traditional solvers often rely on handcrafted heuristics that are difficult to adapt to new problem types. The use of neural networks and reinforcement learning addresses these problems.
- Demonstrates that neural networks can learn their own heuristics for NP-hard problems through trial and error (RL), providing a general-purpose alternative to human-engineered optimization algorithms.

Progress so far

An 8-customer TSP solved using nearest neighbor heuristic (for picking the next node to visit) and A* search (for navigation between nodes) in Albany, NY.

Thank you!